CompTIA

A+ Core 1

Exam 220-1201

A+ 220-1201

Module 3

Installing System Devices

Learning Outcomes

Prepare for A+ Core 1 by:

- Installing and configuring power supplies and cooling
- Selecting and installing storage devices
- Installing and configuring system memory
- Installing and configuring CPUs

Lesson 3.1

Power Supplies and Cooling

Power Supply Units

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- Converts AC to DC
- Outlet voltage compatibility
  - Selector switch or auto

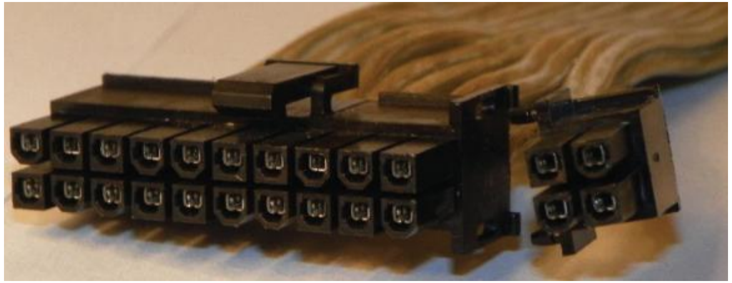
Wattage Rating

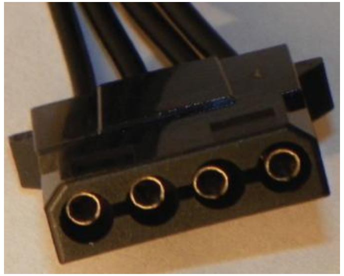
Output Rail (VDC)	Maximum Load (A)	Maximum Output (W)
+3.3	20	130
+5	20	130
+12	33	396
-12	0.8	9.6
+5 (standby)	2.5	12.5

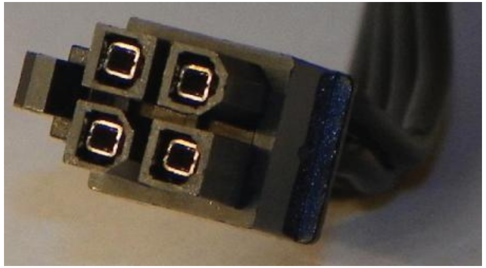
- Match power needs to power supply
  - System instability
  - Component damage
- Energy efficiency
  - 80 plus
    - Bronze
    - Silver
    - Gold

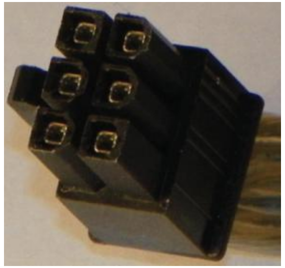
Power Supply Connectors

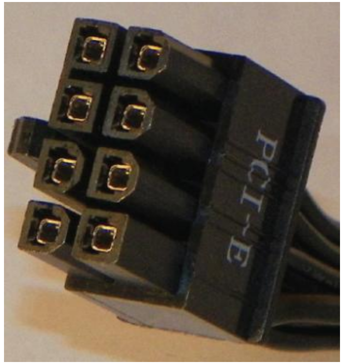
- Transfer power to system components
- P1/24-pin
- Molex
- SATA
- 4/6/8/16-pin connectors

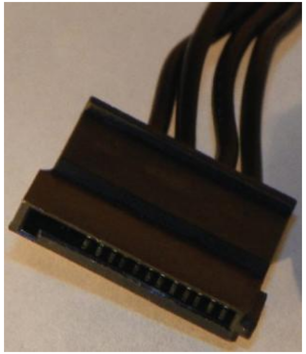












20-pin to 24-pin Motherboard Adapter

- 20-pin legacy
  - Black- Ground
  - Yellow- +12V
  - Red- +5V
  - Orange- +3.3V
- 24 or 20+4 pin

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Modular Power Supplies

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- Detachable cables
  - Customize connectors/cable needs
  - Reduce clutter in chassis

Redundant Power Supplies

Failover protection

Commonly seen in servers and datacenter configurations

Hot swappable vs. cold swappable

Fan Cooling Systems

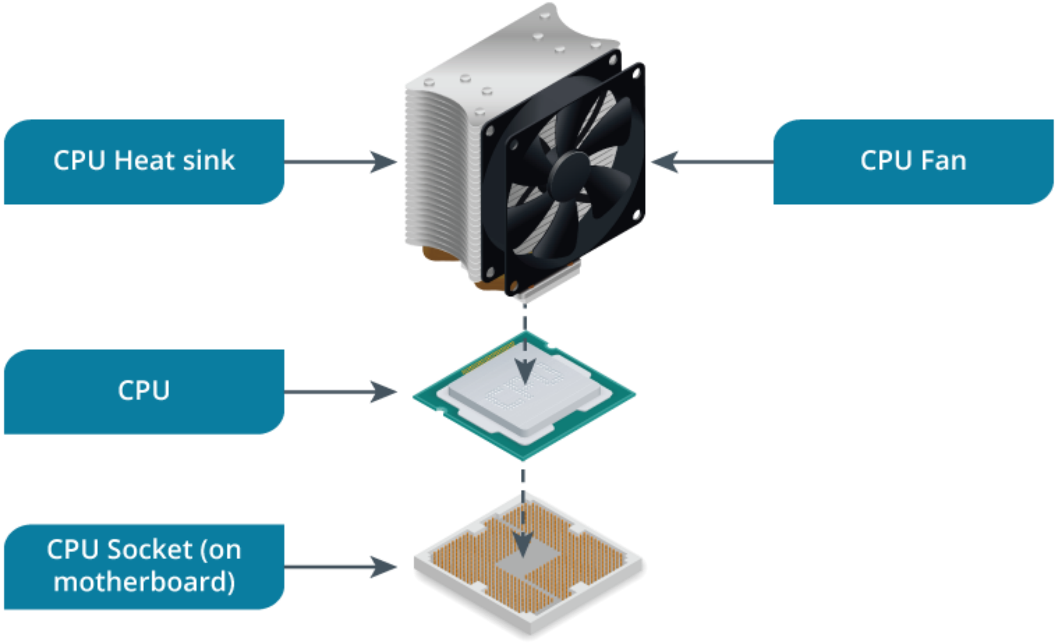
COMPONENTS WORK TOGETHER TO COOL SYSTEM

EXCESS HEAT CAN CAUSE DAMAGE TO COMPONENTS

CONSIDER AIRFLOW PATHWAYS

Heat Sinks and Thermal Paste

- Passive cooling
- Transfer heat better due to surface area
- Bridges microscopic gaps between metal CPU and heat sink block



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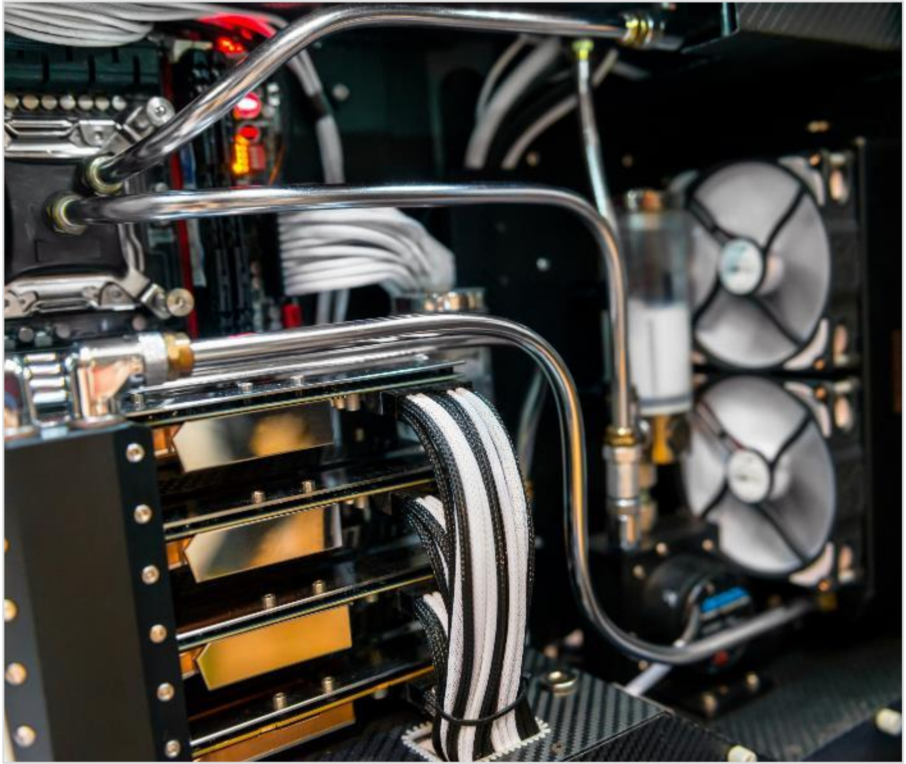
Fans

Active cooling of CPU and components

Air flow patterns

- Cool air from front panel or side panel
- Hot exhaust out rear panel or top

Liquid Cooling Systems



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- Water loop tubing
- Pump
- Water block
- Radiator
- Fan
- Cooling liquid

Lesson Review:

Lab: Install a Power Supply (available in CertMaster Learn and CertMaster Perform)

Lab: Troubleshoot Power Supply Problems (available in CertMaster Learn and CertMaster Perform)

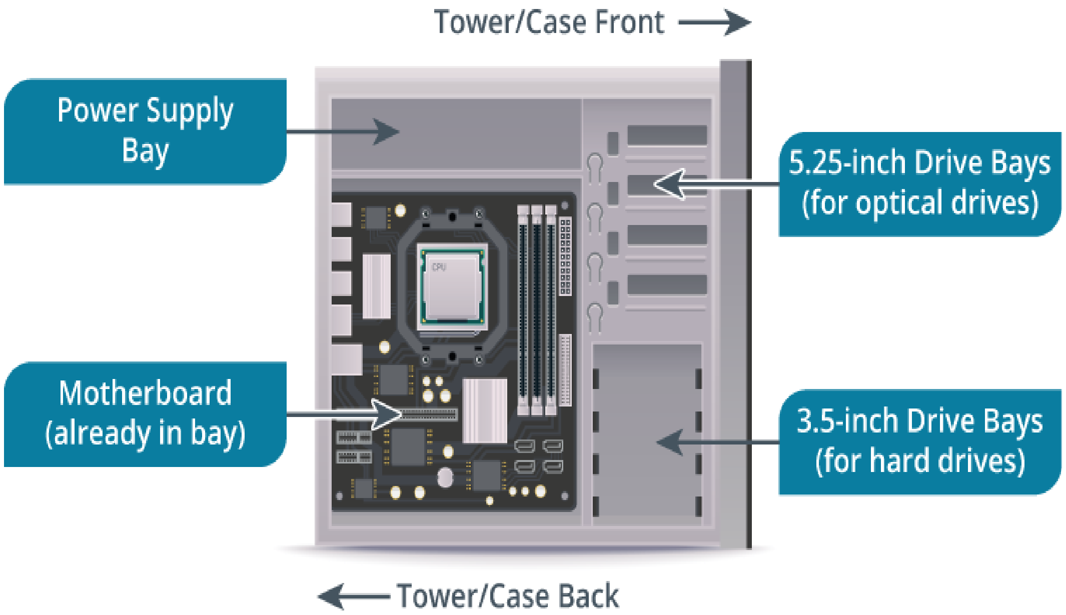
Power Supplies and Cooling

- Power Supply Units
- Wattage Rating
- Power Supply Connectors
- 20-pin to 24-pin Motherboard Adapter
- Modular Power Supplies
- Redundant Power Supplies
- Fan Cooling Systems
- Heat Sinks and Thermal Paste
- Fans
- Liquid Cooling Systems

Lesson 3.2

Storage Devices

Mass Storage Devices



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- Motherboard drive bays
- Removable and externally attached
- Evaluate choice based on:
  - Reliability
  - Performance
  - Use

Solid-State Drives

- No moving components
- Non-magnetic
- High performance
  - Read/write speeds
- Higher cost

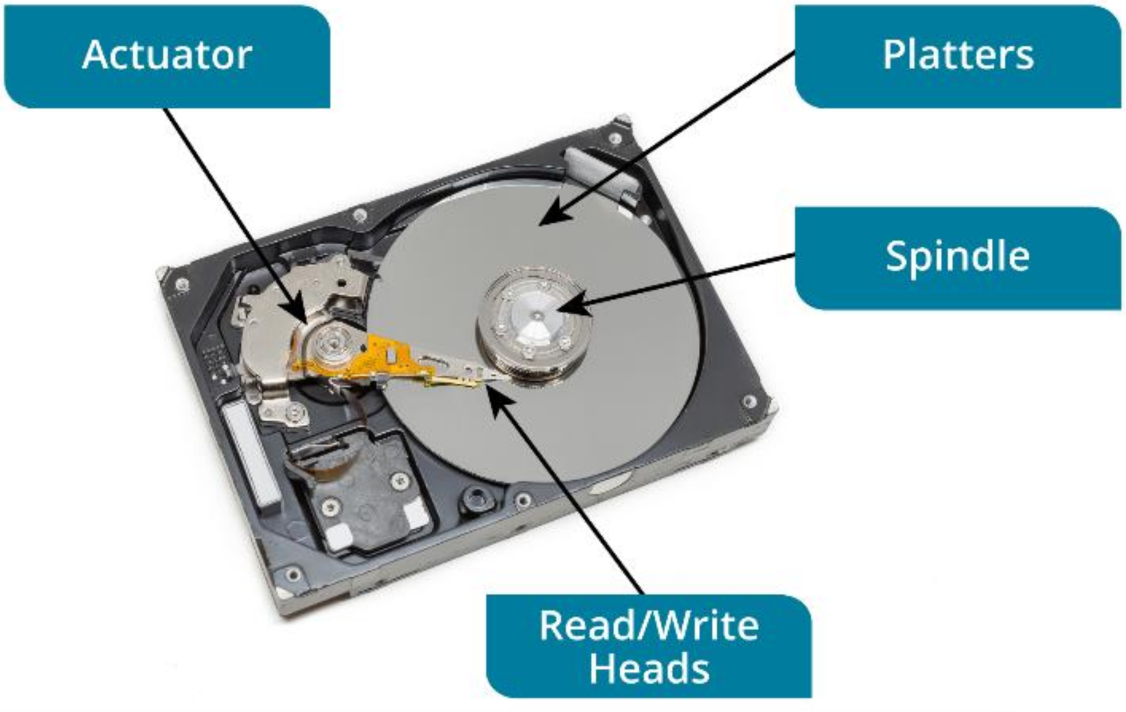
Solid-State Drives

Image © 123RF.com

- Connection methods
  - SATA
  - PCIe- M.2, NVMHCI, or NVMe
  - SAS

Hard Disk Drives

- Magnetic operation
- Lower cost
- Rotational speed
  - 10k or 15k RPM
  - 5400 or 7200 RPM
- SATA interface
- 3.5" or 2.5"



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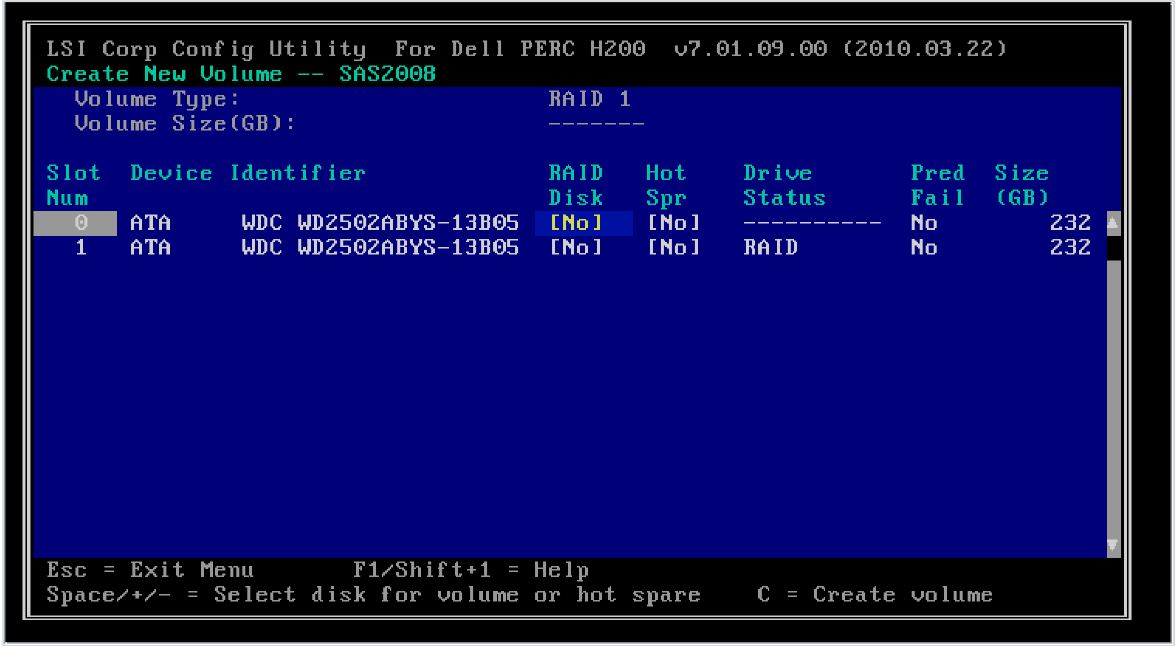
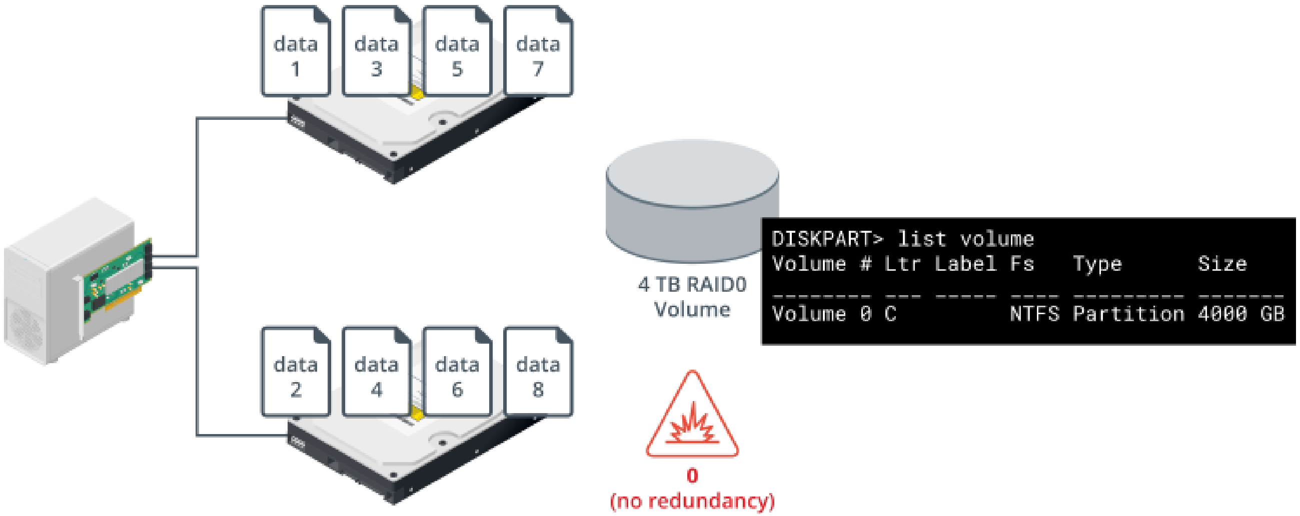
Redundant Array of Independent Disks



- Data distribution across multiple physical drives
- Redundancy and capacity
- RAID levels

RAID 0

- RAID 0- striping without parity
- 2 physical drive minimum
- Better performance than regular hard drive use
- No redundancy



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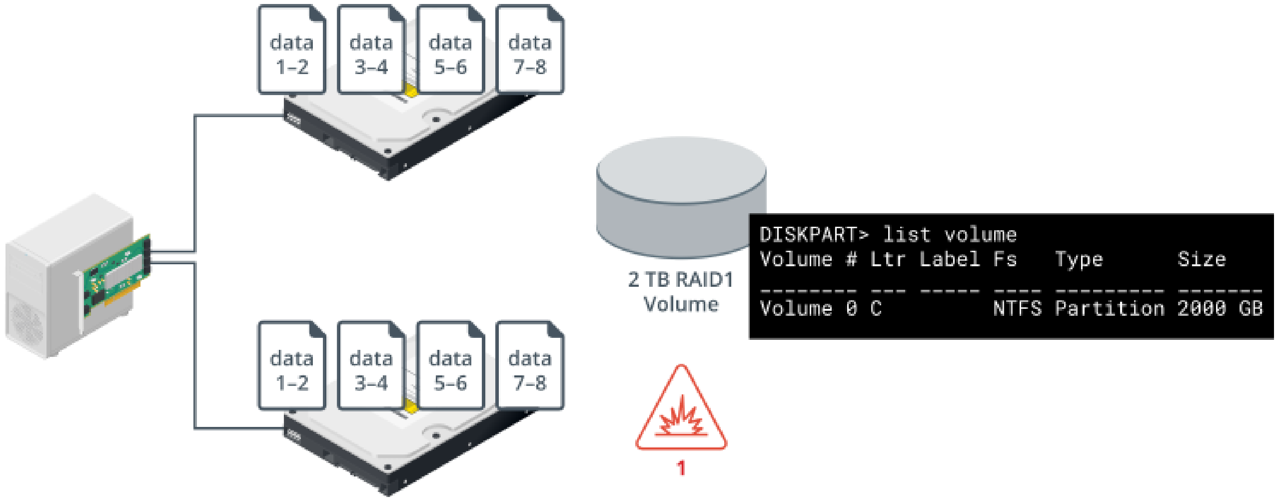
RAID 1

- RAID 1- mirroring
- 2 physical drive minimum
- Redundancy



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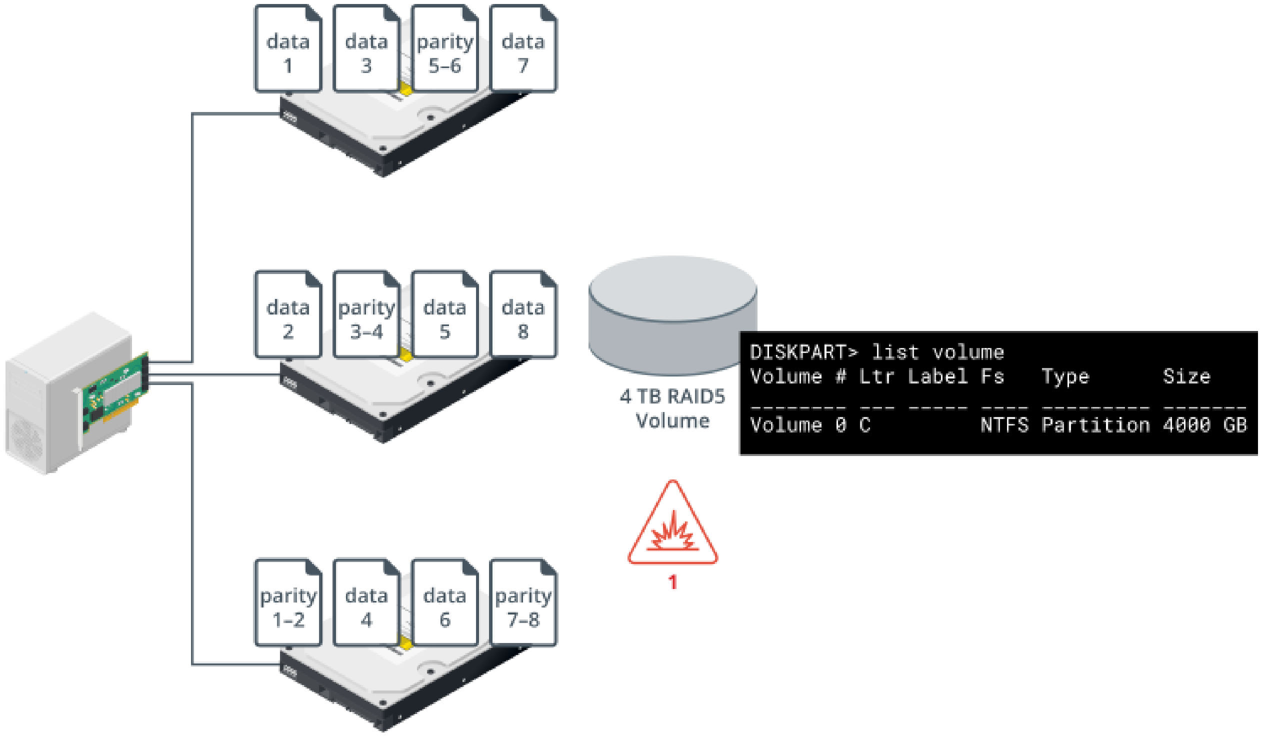
RAID 5



RAID 5- striping with single parity

3 physical drive minimum

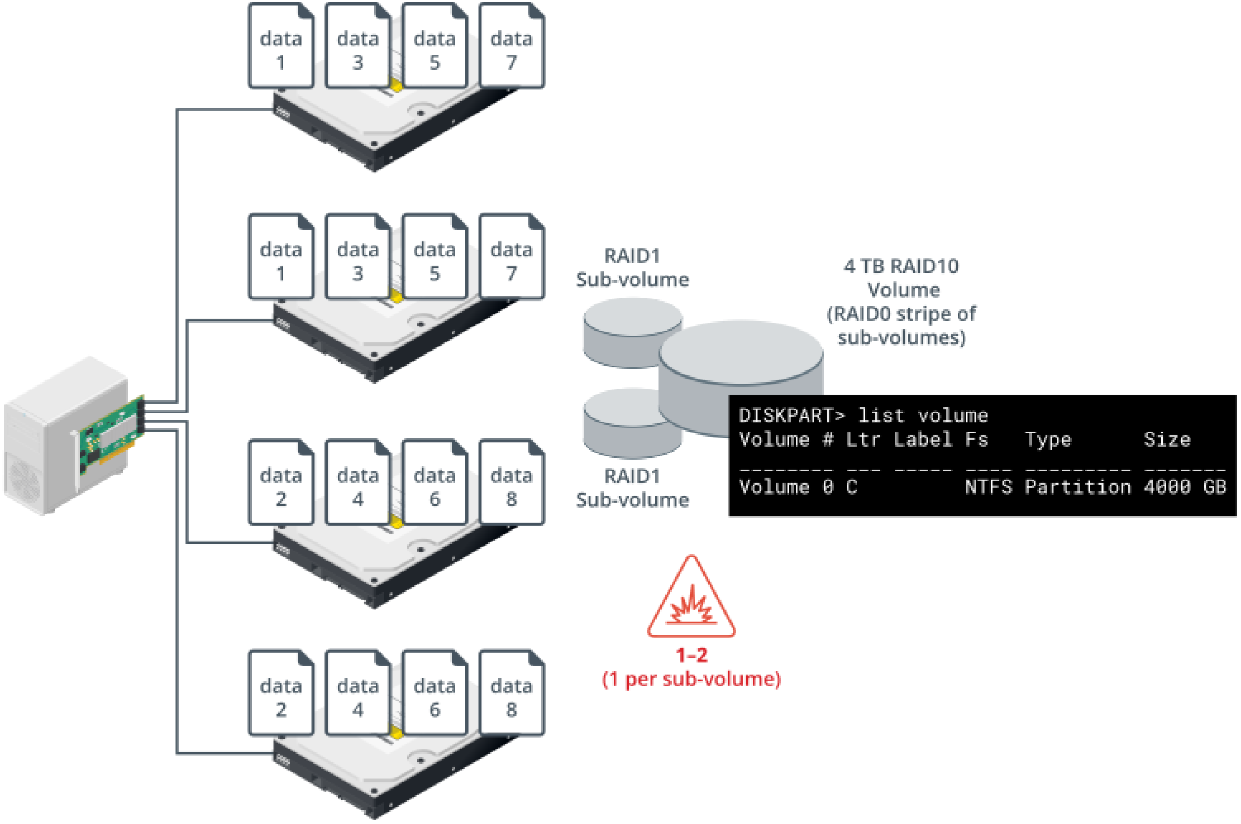
Slower write operation - fast read operation

RAID 10

RAID 10- strip of mirrors

4 physical drive minimum

- Must be an even number of drives



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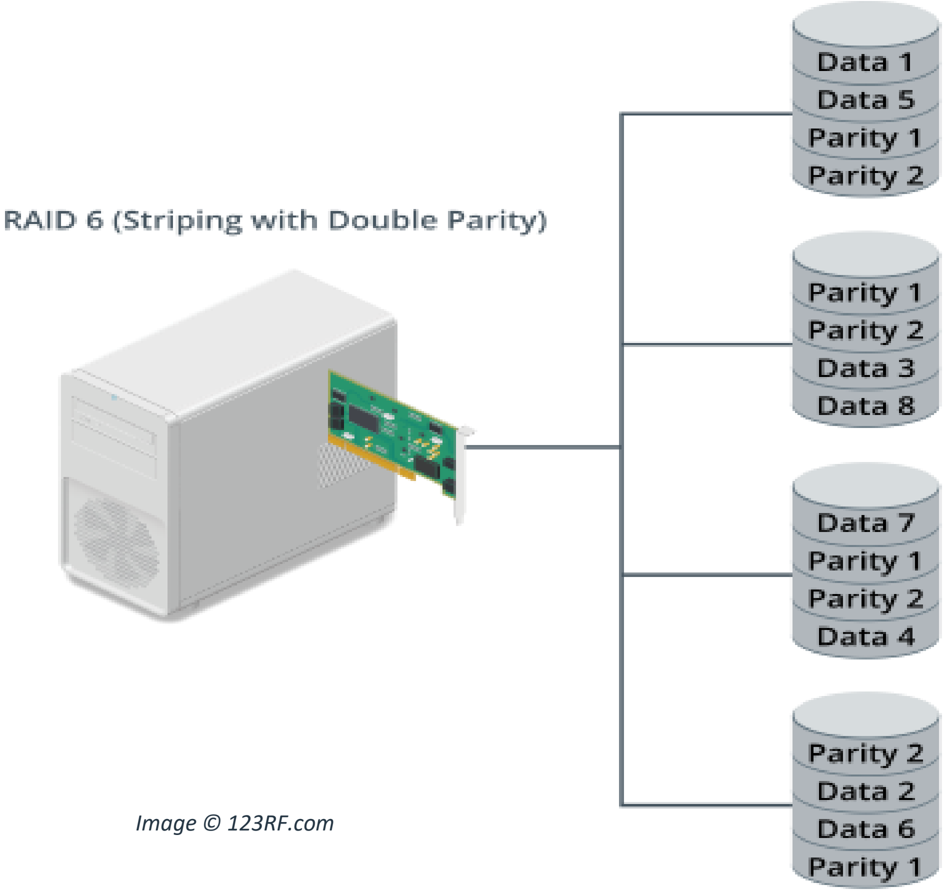
RAID 6 (Striping with Double Parity)



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- Parity is stored in two locations
- 4 physical drive minimum; 2 data/ 2 parity drives
- Better redundancy than RAID 5, but smaller storage space overall
- 2 drive failure will cause degraded performance

Removable Storage Drives

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- Drive enclosures
  - External- USB, Thunderbolt, eSATA
  - Network attached storage

Removable Storage Drives

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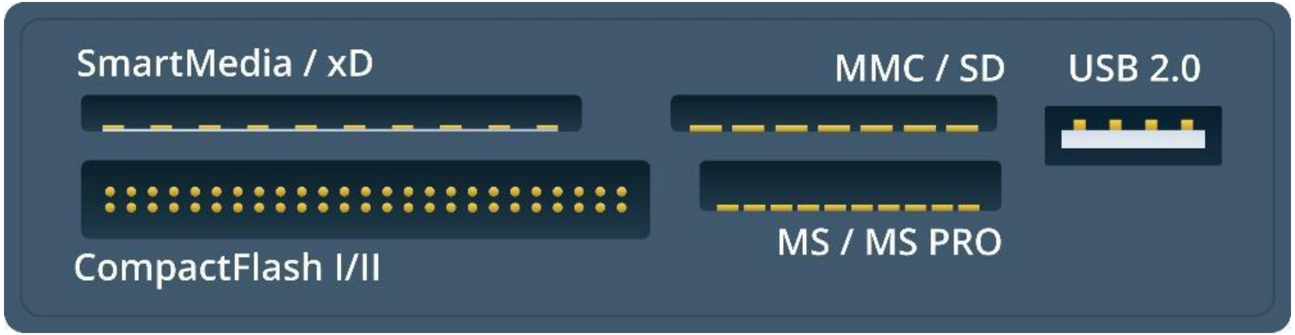
- Flash drives and memory cards
  - USB connector

Removable Storage Drives

- Memory card
  - Card reader for various formats
    - Capacity
    - Speed



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Optical Drives

CD, DVD, Blu-ray

5.25" drive size

Storage size

CD- 700MB

DVD- 4.7GB single layer/sided; 17GB double layer/sided

Blu-ray- 25GB per layer

Transfer rate

Lab Activity

- Lab: Install SATA Devices
  (available in CertMaster Learn and CertMaster Perform)

Storage Devices

- Mass Storage Devices
- Solid-State Drives
- Hard Disk Drives
- Redundant Array of Independent Disks
- RAID 0 and RAID 1
- RAID 5 and RAID 10
- RAID 6 (Striping with Double Parity)
- Removable Storage Drives
- Optical Drives

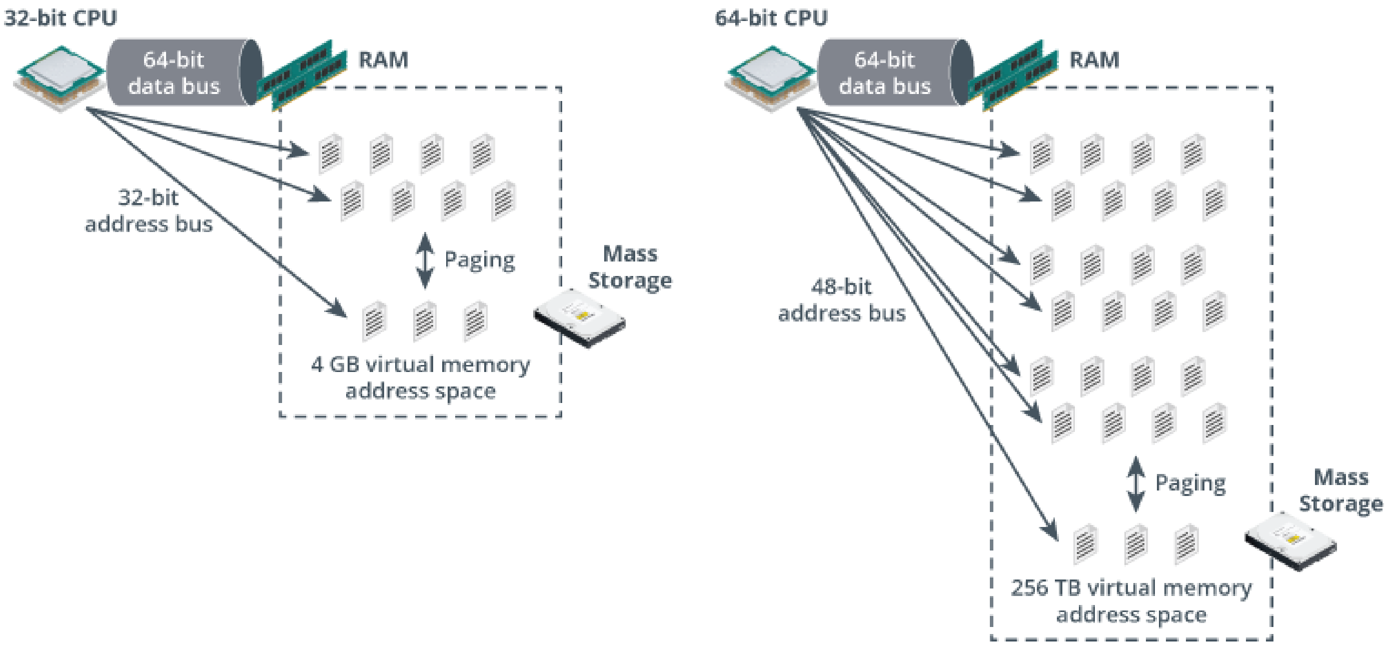
Lesson 3.3

System Memory

System RAM and Virtual Memory (Slide 1 of 2)

- Address space
  - Data pathway
  - Address pathway

System RAM and Virtual Memory (Slide 2 of 2)



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RAM Types (Slide 1 of 2)

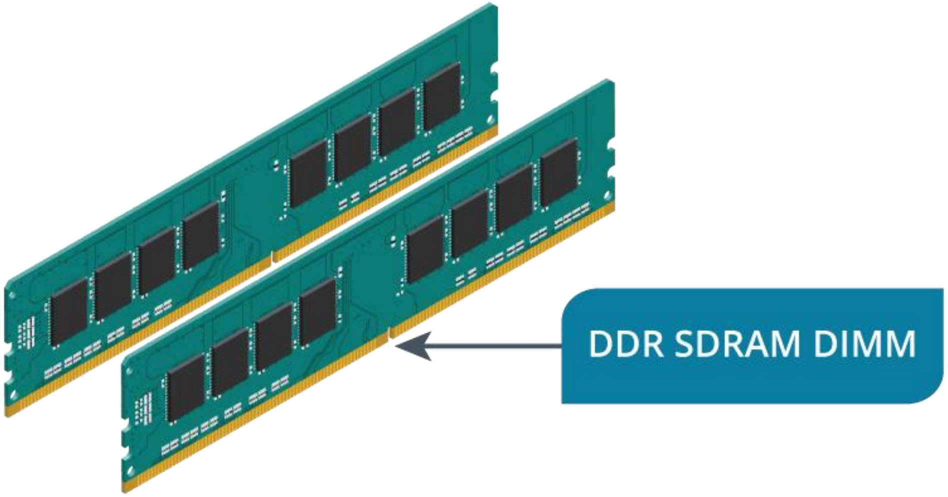
- DRAM
- SDRAM
- DDR SDRAM
- CAS latency

RAM Types (Slide 2 of 2)

RAM Type	Data Rate	Transfer Rate	Maximum Size
DDR1	200 to 400 MT/s	1.6 to 3.2 GB/s	1 GB
DDR2	400 to 1066 MT/s	3.2 to 8.5 GB/s	4 GB
DDR3	800 to 2133 MT/s	6.4 to 17.066 GB/s	16 GB
DDR4	1600 to 3200 MT/s	12.8 to 25.6 GB/s	32 GB
DDR5	4800 up to 8000+	38.4 to 51.2+ GB/s	128 GB or higher

Memory Modules (Slide 1 of 2)

- DIMM
- SODIMM



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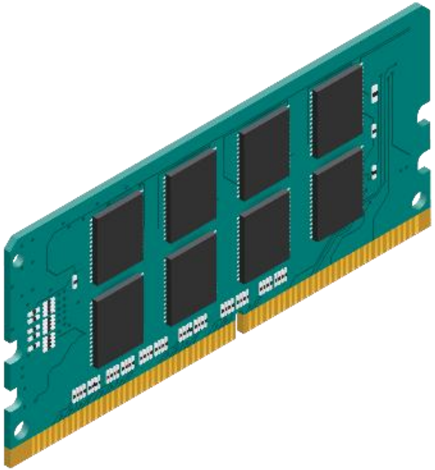
Memory Modules (Slide 2 of 3)



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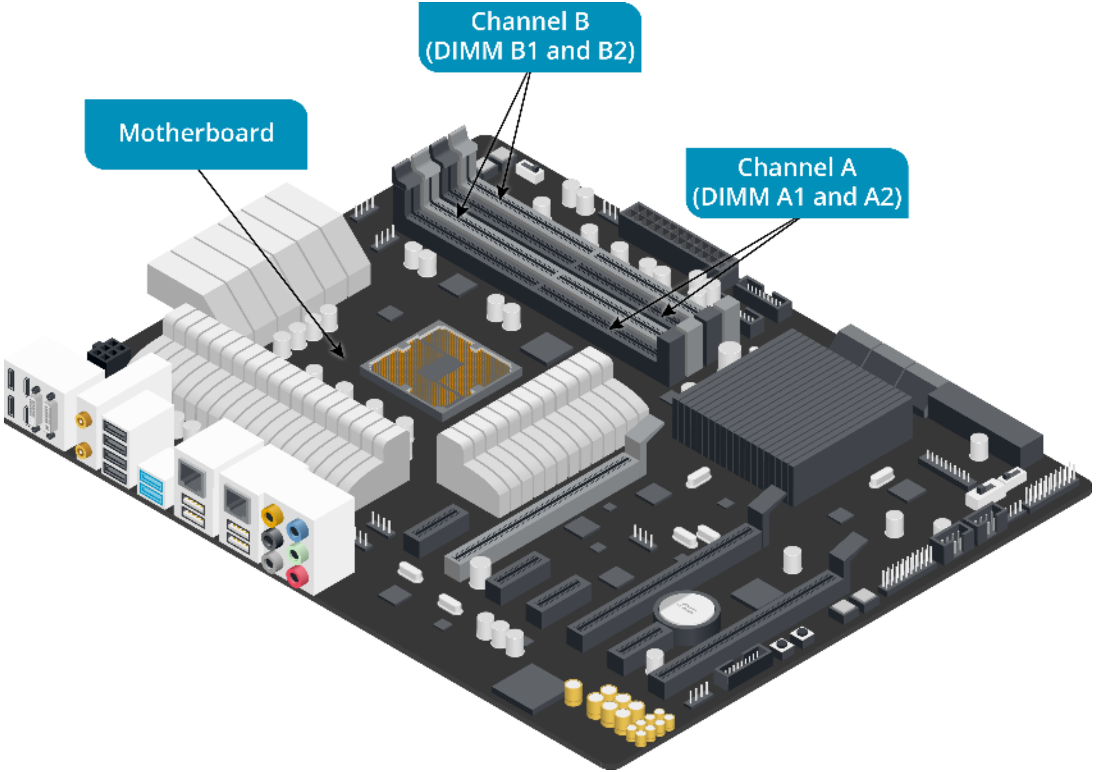
- Installation
- Incompatibility
  - DDR4 module will not fit DDR3 slot

Multi-channel System Memory (Slide 1 of 2)

- Single-channel
- Dual-channel
    - Slot arrangement, installation, setup
- Mismatched module
- Triple/quadruple-channel

Multi-channel System Memory (Slide 2 of 2)



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ECC RAM

Error correction code

Compatibility considerations

- Motherboard and CPU
- DIMM compatibility
- ECC and non-ECC mixing

Lab Activity

- Lab: Select Memory by Sight
  (available in CertMaster Learn and CertMaster Perform)
- Lab: Install Triple Channel Memory
  (available in CertMaster Learn and CertMaster Perform)

System Memory

- System RAM and Virtual Memory
- RAM Types
- Memory Modules
- Multi-channel System Memory
- ECC RAM

Lesson 3.4

CPUs

CPU Architecture

Fetch

Decode

Execute

Write-back

x86 CPU Architecture

- 32-bit instruction set
- Manufacturers
    - Intel
    - AMD
- ALU- Arithmetic Logic Unit
- Cache memory- L1, L2, L3

x64 CPU Architecture

- 64-bit instruction set
- Intel and AMD
- Primary architecture in modern systems
  - Ability to run 32 and 64-bit operating systems and applications

ARM CPU Architecture

- Popular in mobile platforms
- SoC- System-on-Chip
  - Integrated video, sound, network, and storage controllers

CPU Features

- Multithreading
- Symmetric multiprocessing
- Multicore CPUs
- Virtualization

CPU Socket Types (Slide 1 of 2)

- PGA vs LGA
  - AMD- PGA; pins on chip
    - AM4
    - TR4
    - SP3
  - Intel- LGA; pins on socket
    - LGA 1200
    - LGA 1700

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CPU Socket Types (Slide 2 of 2)

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CPU Types and Motherboard Compatibility

- Motherboard socket must match CPU
  - Core configuration
  - Performance
- Desktops
- Workstations
- Servers

Lab Activity

- Challenge Lab: Troubleshoot Memory
  (available in CertMaster Learn and CertMaster Perform)

CPUs

CPU Architecture

x86 CPU Architecture

x64 CPU Architecture

ARM CPU Architecture

CPU Features

CPU Socket Types

CPU Types and Motherboard Compatibility